Cryogenic Detectors in the Direct Detection of Dark Matter

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Abstract

We present a historical and experimental review of dark matter detection, from early dynamical estimates to modern particle-based searches using cryogenic detectors, namely the EDELWEISS and CDMS experiments. This paper outlines the development of dark matter from 19th-century speculation through Fritz Zwicky's analyses of galaxy clusters, Freeman's rotation curve discrepancies, and Rubin's optical spectroscopy of spiral galaxies. These observations support the existence of non-luminous mass and motivate particle dark matter models. We examine leading candidates—including neutrinos, axions, and supersymmetric particles—and briefly evaluate their viability in light of cosmological and laboratory constraints. This paper is chiefly concerned with cryogenic direct detection methods, with a detailed analysis of the EDELWEISS and CDMS experiments. We describe the design, calibration, and data-acquisition strategies for these experiments, specifically improvements in baseline resolution, fiducial-volume definition, trigger efficiency, and the physical detectors. Using data from the 2002 EDELWEISS run, we report a 7.4 kg·d effective exposure and 90% confidence level upper limits on WIMP–nucleon scattering cross sections for WIMP masses below 10 TeV/c^2. CDMS results constrain the parameter space for low-mass WIMPs down to 1 keV recoil energies. These findings exclude portions of the DAMA modulation signal region and may suggest supersymmetric models. We then briefly discuss the effectiveness and future of cryogenic detectors, reinforcing their role in advancing constraints on particle dark matter and viable theoretical models.

1 Introduction

Despite accounting for roughly eighty-five percent of the matter in the universe, dark matter remains among the most pressing mysteries in modern astrophysics. Understanding its properties could reshape both cosmology and particle physics.

Dark matter has evolved from speculation in the 19th century to become one of the central pillars of modern cosmology. Dark matter shapes galaxy formation and dynamics. Its particle nature may also hold clues to the origins of the universe. Tracing the history of the discourse on dark matter highlights how successive advances in observational and theoretical astronomy gradually brought it to the forefront of modern study.

1.1 A Brief History

Father Angelo Secchi was one of the first to mention dark matter (called "dark masses") in 1877 (Bertone and Hooper 2016). Since then, the existence and nature of dark matter have been heavily debated.

With the advent of astronomical photography, scientists noticed the uneven distribution of stars in the sky and questioned whether dark regions were dark because of a low star density or light-absorbing matter (Bertone and Hooper

2016). Arthur Raynard wrote in 1894 that these dark regions seemed to him ”to be undoubtedly dark structures, or absorbing masses in space, which cut out the light from the nebulous or stellar region behind them” (Bertone and Hooper 2016). These speculations were largely fruitless, as there were limitations on the ability to collect persuasive evidence.

Lord Kelvin was a pioneer in dynamically estimating the quantity of dark matter in the Milky Way. He applied statistical mechanical principles to systems of stars: if stars were analogous to gas particles, influenced by gravity, then one can relate the size of the galaxy that they inhabit to the velocity dispersion of the stars (Bertone and Hooper 2016). One could then calculate the approximate mass of the system and compare it with the observed mass using the standard mass-luminosity relation to determine the amount of dark matter (Kelvin 1904). Kelvin found that his predicted velocity dispersion was of the same order as the dispersion that was observed. From this, Henri Poincar´e argued that there was likely a smaller or equal amount of dark matter as there was luminous matter (Bertine and Hooper 2016)

Zwicky (1933) used the virial theorem to estimate the mass of a galaxy cluster (Bertone and Hooper 2016). He estimated the velocity dispersion of Coma cluster galaxies based on the work published in 1931 by Edwin Hubble and Milton Humason on the redshifts of different galaxy clusters (Zwicky 1933, Bertone and Hooper 2016). He calculated the mass of the Coma cluster by multiplying the number of galaxies (800) by the average galactic mass, $10^9\ M_\odot$ as suggested by Hubble (Hubble and Humason 1931). He assumed the system had a size of 10^6 light-years when calculating its potential energy. He then calculated the average kinetic energy of the system and a stellar velocity dispersion of 80 km/s (Zwicky 1933, Bertone and Hooper 2016). The observed radial velocity, however, was roughly 1000 km/s. This discrepancy implies an implausible mass-to-light ratio, unless dark matter is much more abundant than previously thought (Zwicky 1933, Bertone and Hooper 2016). Zwicky established a lower limit of $4.5 \cdot 10^{12}\ M_\odot$ on the mass of the Coma cluster using the virial theorem (1937). He based this on an assumed 1000 galaxies within a $2 \cdot 10^6$ light-year radius and a measured velocity dispersion of 700 km/s. This approach allowed him to solve for the average galactic mass within the cluster. Therefore, Zwicky estimated the average galactic mass in the Coma cluster to be $4.5 \cdot 10^{10}\ M_\odot$ (Zwicky 1937, Bertone and Hooper 2016). Using an estimate of $8.5 \cdot 10^7$ solar luminosities as the average galaxy luminosity in the Coma cluster, he calculated a suspiciously high mass-to-light ratio of roughly 500; this strongly supports the presence of dark matter in the cluster.

This paper (Zwicky 1937) relied on Hubble and Humason (1931), which reported a Hubble constant H_0 = 558 km/s/Mpc. If we use the modern value $H_0 = 67.27 \pm 0.66$ in Zwicky's method, we find a mass-to-light ratio that is much smaller, by a factor of about 558/67.27 (roughly 8.3), than the one Zwicky calculated. Our new mass-to-light ratio remains implausible for the Coma cluster, further reinforcing Zwicky's conclusion that dark matter exists in these galaxies (Bertone and Hooper 2016).

Zwicky's conclusion was supported further by subsequent work on the Virgo

cluster. In this Smith (1936), it was assumed that the outer galaxies follow circular orbits around Virgo, yielding a total cluster mass of $10^{14}\ M_{\odot}$. They subsequently determined the average galactic mass to be $2 \cdot 10^{11}\ M_{\odot}$ by taking the ratio of the total mass to the observed number of galaxies (500). This value greatly exceeds Hubble's estimate of $10^{9}\ M_{\odot}$. Smith (1936) attributed the difference in mass to low luminosity "internebular material" surrounding the galaxies (Bertone and Hooper 2016).

Zwicky (1937) concluded that it was impossible to derive the mass of a galaxy from its rotation curve without additional information, mainly the ellipticity of orbits and the internal viscosity resulting from interstellar interactions (Bertone and Hooper 2016). Chandrasekhar (1942) demonstrated that these interactions are unimportant, enabling the highly accurate approximation that galaxies are systems of non-interacting stars. This allowed the use of galaxy rotation curves to constrain dark matter.

Freeman (1970) was among the first to show that discrepancies between predicted and observed rotation curves require additional mass towards the outer radii of some galaxies (Bertone and Hooper 2016). He compared the expected and observed radii at which the rotation curve of M33 and NGC 300 were expected to peak, and found that the observed radii were larger than expected (Freeman 1970). He concluded that there must be additional matter in these observed galaxies that was undetected (Freeman 1970). Later X-ray observations (Meekins et al. 1971) found that hot intracluster gas in the Pegasus I cluster accounted for less than 2% of the amount needed for gravitational binding (Bertone and Hooper 2016). This limit ruled out gas as a source of the "missing mass" in the velocity dispersion discrepancy (Zwicky 1937) and the rotation curve discrepancy (Freeman 1970). Scientists explored various possible sources of this discrepancy, including HI snowballs, massive collapsed objects, and M8 dwarf stars. However, primordial light element abundances later ruled out these options, instead supporting non-baryonic dark matter (Bertone and Hooper 2016).

Rubin et al. (1978) studied rotation curves in the optical spectra of ten high-luminosity spiral galaxies and found that these curves remained flat even beyond the largest visible radii (Bertone and Hooper 2016). According to Newtonian mechanics,

$$v(r) = \sqrt{\frac{GM(r)}{r}} \tag{1}$$

where G is the gravitational constant, M(r) is the galaxy mass as a function of radius, r is the galaxy radius, and v(r) is the rotational velocity as a function of radius. This suggests that if most of a galaxy's mass is concentrated towards the central bulge, the rotation curve should take a Keplerian form,

$$v(r) \propto \sqrt{1/r} \tag{2}$$

The flat outer rotation curves are convincing evidence for dark matter halos around galaxies.

As a modern extension of Zwicky's work, several methods are currently used to determine galaxy cluster masses, including the virial theorem, weak gravitational lensing, and X-ray emission profiles of hot gas in rich clusters (Bertone, Hooper, and Silk 2024). While observational evidence from astronomy strongly suggested the presence of unseen mass, the question of its physical nature remained open. This shifted the focus from galactic dynamics to particle physics.

1.2 Particle Dark Matter

There are many candidates for dark matter, such as primordial black holes (Carr and Hawking 1974), but by far the most studied candidate in contemporary astronomy is particle dark matter, specifically Weakly Interacting Massive Particles (WIMPs) and Axions (Weinberg 1978).

The standard model (SM) of particle physics has achieved significant success over many years, explaining all observed particles and interactions. However, it is now evident that a more fundamental theory must exist, in which the SM appears in the low-energy limit (Bertone et al. 2024). In the SM, matter is composed of fermions (quarks and leptons), which are massive half-spin particles, and gauge bosons, which are integer-spin particles, govern their interactions (Bertone et al. 2024). Of these bosons, the gluon G_a mediates the strong interactions, and the $W^{\pm}$, Z_0, γ, and Higgs (H^0) bosons mediate the electroweak interactions (Bertone et al. 2024).

Popular non-baryonic dark matter candidates include

- Standard model neutrinos: These were considered good candidates until recently, mainly because of their minuscule interaction cross-section and their known existence. The best constraint on the neutrino mass points to an upper limit of $m_{\upsilon_s} < 2.05$ eV. This upper limit on mass, and therefore density, is much too small to account for all dark matter (Bertone et al. 2024).

- Sterile neutrinos: These particles resemble SM neutrinos but lack SM weak interactions, except mixing (Bertone et al. 2024).

- Axions: These were introduced to solve charge conjugation and parity violation in particle physics (Weinberg 1978). Laboratory searches, stellar cooling, and data from supernova 1987A place an upper limit of roughly 0.01 eV on the mass of axions (Bertone et al. 2024).

- Supersymmetric particles: These are particles such as neutralinos, sneutrinos, and gravitinos from supersymmetric models of particle physics. Neutralinos are by far the most studied (Bertone et al. 2024).

There remain many other novel candidates for dark matter (Bertone et al. 2024), but they remain beyond the scope of this review.

1.3 Current Direct Detection Experiment

Direct detection experiments for dark matter are laboratory experiments to measure its mass and scattering cross-section (Bertone and Hooper 2016). Cryogenic detectors detect phonon heat and ionization responses to particle interactions. This paper reviews the current state of cryogenic detector experiments, focusing on methods, empirical results, and the limits they place on the physical properties of dark matter. As cryogenic detector methods and technology advance, we can set more stringent limits on the mass and scattering cross-section of WIMP dark matter candidates.

This paper will focus on two cryogenic detector experiments: EDELWEISS, which placed an upper limit on the scattering cross-sections for WIMP mass(M_{wimp}) < 10 TeV/c^2 with a 90% confidence level (Benoit et al. 2002), and CDMS, which excluded parameter space for 3 GeV/c^2 < M_{wimp} < 4GeV/c^2 (Akerib et al. 2011). This paper will describe the methods of these experiments, their results, the differences between the experiments, and summary conclusions for each experiment.

2 Methods

We will first cover the experimental methods of Cryogenic detectors, namely EDELWEISS and CDMS.

2.1 EDELWEISS

This experiment was conducted in the Laboratoire Souterrain de Modane under a 4800 water-equivalent rock overburden (Benoit et al. 2002). To reliably detect WIMPs, which have a very low interaction rate (<1 event/d/kg), it is necessary to maximize the mass of each detector (Navick et al. 2000). However, increasing this mass weakens heat signals due to the greater heat capacity. It may also hinder charge collection through space-charge accumulation or trapping (Navick et al. 2000). To overcome this challenge, Navick et al. (2000) introduced 320 g germanium ionization-heat detectors that were significantly larger than earlier versions. Their increased mass enhances the likelihood of detecting WIMP interactions (Navick et al. 2000). Each detector contains a thermal sensor and an absorber, which is made from a high-purity germanium monocrystal measuring 70 mm in diameter and 20 mm thick (Navick et al. 2000).

The flat surfaces and chamfers of the crystals have been metallized using two distinct techniques to facilitate ionization measurement (Navick et al. 2000). In the first method, used for the GeAl-5 and GeAl-6 detectors, electrodes were created by sputtering a 100 nm aluminium layer onto the crystal surface following plasma etching (Navick et al. 2000). The second technique, applied to the Ge-8 detectors, formed electrodes by implanting boron ions on one side and phosphorus on the opposite side (Navick et al. 2000). A 2 mm passivation-free edge separates these electrodes, enhancing the electric field and

reducing charge-carrier trapping (Navick et al. 2000). The GeAl-6 detector features a guard ring that is electrically isolated from the central region, enabling the rejection edge events through radial localization (Navick et al. 2000).

The thermal sensors in these detectors consist of neutron-transmutation-doped (NTD) germanium crystals with volumes ranging from 1.5 mm to 3 mm (Navick et al. 2000). These crystals also have a surface, with area between 1.7 and 4.2 mm, that exchanges heat with the absorber (Navick et al. 2000). The heat sensors are attached to two small surfaces on a crystal chamfer and coated with a 100 nm thick, 8 mm diameter gold layer (Navick et al. 2000). The detectors are mounted on a compact holder with 1 to 1.5 mm spacing, which helps reject coincident detections in different bolometers (Navick et al. 2000). Additionally, this arrangement provides passive shielding, particularly against beta radiation and low-energy X-rays (Navick et al. 2000).

The detectors utilize various thermal links (Navick et al., 2000). Specifically, Ge-8 is equipped with four gold wires, GeAl-5 with a single copper wire, and GeAl-6 with three copper wires. These thermal links are attached to the same gold layer as the NTD germanium with epoxy resin. At the opposite end, the wires are either bonded or glued to copper strips (Navick et al. 2000). Ionization and heat measurements were performed via aluminium wires affixed to thin gold layers on both the absorber and the NTD electrodes (Navick et al. 2000).

The 2000 EDELWEISS experiment used three of these 320 g germanium bolometers (Julliard et al. 2000). These bolometers featured aluminium electrodes, which helped minimize charge collection issues from surface events compared to implanted electrodes (Julliard et al. 2000). As previously mentioned, a guard ring was used to identify signals originating near the detector edges (Julliard et al. 2000). Both the central and guard ring electrodes were polarized at 6.3 V (Julliard et al. 2000). Throughout the experiment, linear cross-talk of about ten percent was observed between the two channels, but this was corrected by simultaneously analyzing the signals from both electrodes (Julliard et al. 2000).

A fiducial volume—about 50% of the total detector volume, as determined by neutron calibration—was defined by selecting events that produced significant amplitude only on the central electrode (Julliard et al. 2000). Potential dark matter interactions were tracked by measuring temperature changes with a DC-polarized NTD germanium sensor possessing a resistance of approximately 4 MΩ (Julliard et al. 2000). The temperature was kept stable within 10 μK of 27 mK (Julliard et al. 2000).

In the 2000 experiment, the ionization channel for the central zone had a baseline resolution of 2 keV full-width at half-maximum (FWHM) and a resolution of 3 keV at 122 keV, with a detection threshold of about 6 keV electron equivalent—constrained mostly by microphonics. For the guard ring, the baseline was 1.5 keV, and the resolution at 122 keV was 2 keV, with a detection threshold around 5 keV electron equivalent (Julliard et al. 2000). The heat channel had a baseline resolution of 1.5 keV and a resolution of 3 keV at 122 keV. The initial trigger rate, before data filtering, was a few events per minute

(Julliard et al. 2000).

In the 2002 EDELWEISS experiment, three 320 g germanium detectors—each measuring 70 mm in diameter and 20 mm in height—were operated simultaneously (Benoit et al. 2002). As before, each detector featured a segmented electrode to define the central region and guard ring (Benoit et al. 2002). To enhance reliability, all electrical contacts were ultrasound-bonded to the electrodes (Benoit et al. 2002), replacing the previous method of gluing Navick et al. 2002).

Two EDELWEISS detectors, GeAl-9 and GeAl-10, closely resembled the GeAl-6 detector described by Navick et al. (Benoit et al. 2002). In contrast, the third detector, GGA1, featured a 60 nm hydrogenated amorphous germanium layer beneath the 70 nm aluminium electrodes and on all exposed surfaces (Benoit et al. 2002).

In the 2002 experiment, the authors increased the thermalization of the EDELWEISS detectors and the size of the NTD heat sensors for a better sensitivity (Benoit et al. 2002). As a result, these authors were able to operate the detector at 17 mK regulated to within 10 μK

The data collection system for the EDELWEISS 2002 experiment employed a fully digitized ionization trigger (Benoit et al. 2002). Signals from the six ionization channels and three heat channels were continuously digitized at sampling rates of 2 ksample/s and 200 ksample/s, respectively (Benoit et al. 2002). These digitized signals were transmitted to the data-collection computer through an optical link (Benoit et al. 2002). Ionization data were filtered below 1200 Hz using a fourth-order infinite-impulse-response high-pass elliptic filter to suppress most microphonic noise (Benoit et al. 2002). The trigger was set by imposing a minimum threshold on the amplitude of any filtered ionization channel (Benoit et al. 2002).

To detect WIMPs, it is essential to eliminate signals from unrelated external sources (Benoit et al. 2002). In the EDELWEISS 2002 experiment, the authors calibrated ionization and heat responses to gamma rays using Cobalt-57 and Cobalt-60 sources (Benoit et al. 2002). The performance of the GeAl-6 detector in the 2000 experiment was constrained by poor ionization and heat channel baseline resolution (Benoit et al. 2000, Benoit et al. 2002). In the 2002 experiment, Benoit et al. maintained the 122 keV resolution of the new detectors at a level similar to that of the GeAl-6 detector used in Benoit et al. (2000), but significantly improved the baseline resolution. All ionization baseline resolutions in the 2002 experiment were under 1.5 keV FWHM, while the heat channel baseline resolutions were 1.3, 0.5, and 0.4 keV for GGA1, GeAl-9, and GeAl-10, respectively (Benoit et al. 2002).

The improved baseline resolution was calibrated using decays from cosmic-ray-induced long-lived isotopes zinc-64 and germanium-68, as illustrated in Fig. 1 of Benoit et al. (2002). The 8.98 and 10.37 keV spectral lines—corresponding to zinc and germanium isotope decays, respectively—are distinctly resolved in GeAl-9 and GeAl-10, achieving a 0.6 keV FWHM resolution. In GGA1, the resolution is slightly lower at 1.2 keV FWHM; however, the two-peak structure remains visible (Benoit et al. 2002). The authors attributed GGA1's reduced

resolution to the smaller volume of its NTD sensor and greater susceptibility of its center electrode ionization channel to microphonics (Benoit et al. 2002).

Benoit et al. (2002) determined the ionization trigger threshold using two distinct methods. In the first approach, they extracted the threshold corresponding to 50% efficiency by fitting the low-energy region of the Compton plateau, measured with a cobalt-60 source. In the second method, they employed a californium-252 neutron source and leveraged the fact that neutron scattering creates numerous coincident events involving two or more detectors. To construct the efficiency curve, the authors compared ionization energy distributions, including and excluding a detector in the trigger pattern. Events triggered by at least one additional detector served as the reference group (Benoit et al. 2002).

Both methods yielded ionization threshold measurements that were consistent within 0.2 keV (Benoit et al. 2002). The threshold values for GGA1 corresponding to an efficiency of fifty percent were 3.7±0.2 for the Compton method and 3.5±0.1 for the neutron scattering method. These showed significant improvement over the GeAl-6 threshold values of Benoit et al. (2000) (Benoit et al. 2002). The corresponding values for the GeAl-6 detector varied between 5.7 and 11 keV (Benoit et al. 2000) and constrained the authors' analysis to recoil energies greater than 30 keV (Benoit et al. 2002)

In the 2002 EDELWEISS run, the authors defined the quenching factors Q as the distribution of ratios of ionization signals to recoil energies. These Q were recorded in the presence of a gamma-ray source (Benoit et al. 2002). This revealed problems with charge collection in the detectors GeAl-9 and GeAl-10 (Benoit et al. 2002). Figure 2 from Benoit et al. (2002) displays the Q distributions for the different detectors during the 2002 run for nuclear recoil energies from 20 to 200 keV, recorded in the presence of cobalt-57 and cobalt-60 sources. Because GeAl-9 and GeAl-10 show similar behavior, their distributions were added together, and all detector distributions were normalized to the GGA1 entry count (Benoit et al. 2002). While the GGA1 distribution is centered about one, as expected by construction, the combined GeAl-9 and GeAl-10 distribution shows a narrower peak centered at one due to their higher heat resolution (Benoit et al. 2002).

The relatively flat tail of events in Figure 2 of Benoit et al. (2002) is problematic. While 1.3 and 2.2 percent of events have values of Q below 0.5 in the GeAl-9 and GeAl-10 detectors, respectively, approximately 0.01 percent of events have a Q below 0.5 in GGA1 (Benoit et al. 2002). Based on the 2002 cobalt-57 and cobalt-60 calibration runs, the authors expected that the tails would produce false positives at several events/kg/d in GeAl-9 and GeAl-10, which was later verified (Benoit et al. 2002). The authors therefore decided to use only GGA1 data to derive a limit on WIMP interactions (Benoit et al. 2002). The authors posit that the reduction of charge-collection issues in GGA1 could indicate that the hydrogenated amorphous germanium layer helped mitigate them.

Benoit et al. (2002) defined a fiducial volume to exclude events occurring in the outer perimeter of the detector as in Julliard et al. (2000). This prevents

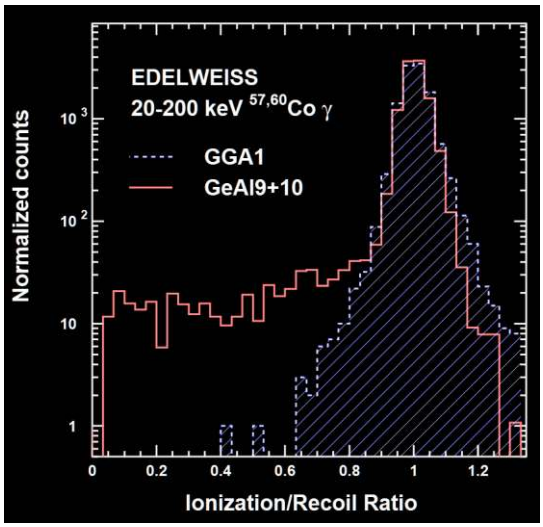

Figure 1: Q distributions obtained from cobalt-57 and cobalt-60 sources. The orange solid line represents the GeAl9 and GeAl10 combined distribution, while the blue dashed line represents the GGA1 distribution (Figure 2 from Benoit et al. 2002).

false positives from external radioactivity and suppresses charge collection issues, which are more common in the outer perimeter (Benoit et al. 2002). These authors used two methods to define the fiducial volume in the detectors: the first uses a californium-252 source to measure the fraction of nuclear recoil events, which is then compared to a Monte Carlo simulation of neutron interactions in the detector (Benoit et al. 2002). The second method uses the fiducial acceptance, defined by ranges in Q and recoil energy, to select a fiducial volume fraction from the total intensity of uniformly decaying zinc-65 (8.98 keV peak) and germanium-68 (10.37 keV peak). These methods gave identical results for the GGA1 detector (57 ± 3%) and yielded similar results for the other detectors (within 3 %) (Benoit et al. 2002). These results set the lower bound of nuclear recoil selection energy to 20 keV and they set the upper bounds to 33 keV (M_{wimp} = 20 GeV/c^2), 86 keV (M_{wimp} = 100 GeV/c^2), and roughly 110 keV (M_{wimp} > 10 TeV/c^2) (Benoit et al. 2002). This allowed the band-detection efficiency to be as uniform as possible and excluded regions where gamma-ray rejection was expected to be worse than 99.9% (Benoit et al. 2002). The fiducial acceptance for WIMP interactions thus includes 90% to 95% of all recoils above 20 keV.

The low-background dataset includes all physics runs collected between February and May 2002 (Benoit et al. 2002). Data-taking began several months after the detectors were installed at the experiment site, following calibration, resolution optimization, and extended exposure to a cobalt-60 source with all electrodes shorted. The authors kept the data collection conditions as homo-

geneous as possible until an accidental detector warm-up interrupted the run. Stability was monitored continuously and further verified through weekly cobalt-57 calibrations and two neutron calibrations. In total, the low-background run lasted 54 days. Of this, 2.0% was lost because of regular electrode shorting to prevent buildup of space-charge, 3.9% due to data-acquisition dead time, and 6.2% during several episodes of excessive microphonics noise, which degraded baseline resolution. The overall exposure amounts to 15.1 kg·d, with 8.6 kg·d in the central fiducial volume. After correcting for nuclear recoil band acceptance, the effective exposure was 7.4 kg·d.

2.2 CDMS

Akerib et al. (2011) reviews the methods and data from the last run of the first six CDMS II detectors at the Stanford Underground Facility (SUF) (Akerib et al. 2003). The goal of this experiment was to extend the lowest-energy recoils to roughly 1 keV for germanium targets and to 2 keV for silicon targets. The SUF setup was heavily shielded against external radiation sources with approximately 17 meters water-equivalent overburden, which nearly eliminated interference from cosmic rays and cut muon flux by 80% (Akerib et al. 2011). Muon-detector interactions from the remaining muon flux were identified by a high-efficiency hermetic plastic scintillator muon shield, which enclosed several layers of passive shielding (Akerib et al. 2011). Surrounding the detector cold volume was a 15 cm-thick lead shield and a 25 cm-thick polyethylene shield (Akerib et al. 2011). The polyethylene shield was used to degrade external neutrons, and the lead shield was used to attenuate external photons (Akerib et al. 2011). The detector was further directly surrounded by 11 kg of polyethylene and 1 cm of ancient lead inside the 20 mK cold volume, which was enclosed by radiopure copper walls (Akerib et al. 2011). At the core of the configuration, six Z-sensitive Ionization and Phonon-mediated (ZIP) detectors (Irwin et al. 1995) were positioned in a vertical stack, with each neighboring detector spaced 2.2 mm apart with no material in between them. Table 1 of Akerib et al. (2011) lists the detector names, materials, and masses. They are listed by position in the detector tower from top to bottom.

Name	Material	Mass (grams)
Z1	Ge	230.5
Z2	Ge	227.6
Z3	Ge	219.3
Z4	Si	104.6
Z5	Ge	219.3
Z6	Si	104.6

Table 1: CDMS II detectors listed by position from top to bottom in the detector tower, including name, material, and mass (Table I from Akerib et al. 2011)

All detectors have, on their bottom surfaces, a circular inner electrode (q-inner) and a concentric outer electrode (q-outer) that identifies and rejects energy depositions in the outer detector volume (Akerib et al. 2011). On the top surface of each germanium detector, 4144 Al and W superconducting Quasiparticle-trap-assisted Electrothermal-feedback Transition-edge sensors (QETs) are photolithographed (Akerib et al. 2011). On the top surface of each silicon detector, 3552 of these QETs are photolithographed (Akerib et al. 2011). In each quadrant, 1036 (Ge) or 888 (Si) QETs are electrically connected, forming four independent phonon sensors divided by orthogonal boundaries (Akerib et al. 2011). By comparing pulse-height and timing differences between these sensors, it is possible to reconstruct the positions of events in the plane of the top (and bottom) surface (Akerib et al. 2011).

After SQUID array and room-temperature electronics amplification, the four phonon signals from each detector were duplicated twice (Akerib et al. 2011). One set was band-pass filtered and summed to form the triggering phonon energy. If this triggering energy surpassed a specified discriminator threshold, the resulting logical pulses from all detectors were OR'ed to form the experimental trigger (Akerib et al. 2011). The second set was digitized, optimally filtered in software, and summed to yield the reconstructed phonon energy (Akerib et al. 2011). The hardware bandpass filter poles were chosen to approximate the software filtering (Akerib et al. 2011). Discriminator thresholds were tuned to the lowest possible levels, maintaining a $\lesssim$1 Hz trigger rate during WIMP searches, to ensure that the data were dominated by genuine particle interactions with only occasional noise-induced events (Akerib et al. 2011).

Recoil energy deposition generates two signals in the detectors: spectra of high-frequency phonons and electron-hole pairs (Akerib et al. 2011). The mean energy per pair is ϵ = 3.0 eV in Ge and ϵ = 3.8 eV in Si (Akerib et al. 2011). Ionization was measured by grounding the QET side and applying a -3 V to -6 V bias to the opposing electrodes, driving free electrons (holes) towards the QET electrode (Akerib et al. 2011). The electrode signals were amplified by charge-sensitive amplifiers with field-effect transistor front ends, allowing the ionization energy Q to be inferred from the total free-ionization (Akerib et al. 2011). The total phonon signal consists of recoil-produced phonons (primary phonons), drift heat phonons from drifting charge, and phonons produced upon recombination of holes and free electrons, typically at detector surfaces (Akerib et al. 2011). From these, the QETs allowed the calculation of the total phonon energy P_{total}. The recoil energy, E_{recoil}, equals the sum of energies from recombination phonons and primary phonons, obtained by subtracting the drift heat phonons:

$$E_{recoil} = P_{total} - (V \cdot e/\epsilon)Q \tag{3}$$

where V is the magnitude of the bias voltage, e is the charge of the proton, ϵ is the mean energy per pair created, and Q is the ionization energy (Akerib et al. 2011).

CDMS detectors simultaneously measure interaction event ionizations and

phonons to distinguish WIMP candidate nuclear recoils from the dominant background of photon-induced electron recoils (Akerib et al. 2011). The ionization yield is defined as

$$Y = Q/E_{recoil} \tag{4}$$

where Q is the ionization energy. The ionization yield is higher for electron recoils, enabling nearly perfect discrimination for each event for recoil energies above 10 keV (Akerib et al. 2011). At lower energies, however, electronic noise broadens the yield distribution, reducing discrimination below 10 keV and effectively eliminating it below roughly 2 keV (Akerib et al. 2011).

The authors used electron recoils from gamma-ray sources to calibrate the Q and E_{recoil} energy scale, fixing the electron-recoil ionization yield to unity by construction (Akerib et al. 2011). Nuclear recoils from a neutron source were used to determine the dependence of the nuclear recoil ionization yield on recoil energy. This nuclear recoil yield rises smoothly from roughly 0.2 at E_{recoil} = 2 keV to roughly 0.4 at E_{recoil} = 100 keV (Akerib et al. 2011).

Direct measurement of nuclear recoil ionization yield below 2 keV with ZIP detectors is challenging. However, Ge detector measurements have been performed accurately down to roughly 0.25 keV (Akerib et al. 2011). With these measurements, Akerib et al. (2011) extrapolated a power law from 2 keV to lower energies, where the yield vanishes at 0 keV. This extrapolation was only applied to estimate the energy dependence of several detection efficiencies at the lowest recoil energies. At such low energies, E_{recoil} is slightly less than P_{total}, so the correction to the extrapolation is small and its associated uncertainty is also small relative to other sources of uncertainty (Akerib et al. 2011).

One approach to estimating recoil energy uses both the measured Q and P_{total} in Eq. 3 for each event (Akerib et al. 2011). This method includes uncertainties from both quantities in an estimate for E_{recoil} and introduces correlations with the ionization yield (Akerib et al. 2011). This is the traditional CDMS procedure, which remains accurate regardless of whether the event is an electron or a nuclear recoil (Akerib et al. 2011). Analysis cuts and efficiencies in Akerib et al. (2011) employ this event-level estimate, which is referred to as the Q-corrected recoil energy. Unless otherwise noted, all recoil energies from Akerib et al. (2011) are Q-corrected.

One can also estimate the recoil energy from P_{total} alone by scaling it to resemble the average ionization yield response determined from calibration data (Akerib et al. 2011). Since electron recoils have an average ionization yield of one, their ionization energy equals their recoil energy (Akerib et al. 2011). Substituting Q in for E_{recoil} in Eq. 3 shows that E_{recoil} equals half of P_{total} for Ge ZIP-detectors operated at -3 V, and that E_{recoil} equals one third of P_{total} Ge ZIP-detectors operated at -6 V (Akerib et al. 2011). This estimate, called the Y_{ER}-corrected recoil energy, provides higher resolution and is particularly helpful for characterizing ZIP-detector responses to gamma-ray and x-ray sources (Akerib et al. 2011).

One can make an analogous recoil-energy estimate for nuclear recoils by subtracting the energy-dependent drift heat, as determined from the average

ionization yield in californium-252 calibrations (Akerib et al. 2011). This Y_{NR}-corrected recoil energy benefits from incorporating electronic noise from the phonon channel alone (Akerib et al. 2011). Both hardware and software energy thresholds were solely dependent on the total phonon signal, so the corresponding threshold efficiencies for nuclear recoils are expressed in terms of the Y_{NR}-corrected recoil energy (Akerib et al. 2011).

The first CDMS II ZIP tower WIMP-search data set, called Run 21, was collected from December 2001 to June 2002 (Akerib et al. 2002). The Ge and Si detectors were operated at -3 V bias voltages for 94 days, yielding roughly 66 live days of 3 V data. The authors analyzed the 3 V dataset using a 5 keV threshold, yielding one of the strongest exclusion limits on the spin-independent WIMP-nucleon cross section for low-mass WIMPs (Abrams et al. 2002, Akerib et al. 2011). The authors acquired an additional 52 live days of data with a -6 V bias to reduce surface event background (Akerib et al. 2011). While the higher bias increased the collection of charge near detector surfaces, it impaired discrimination of phonon pulse rise time, leading to slightly more surface events leaking into the signal region (Akerib et al. 2011). Later CDMS operations used the lower bias, but Akerib et al. (2011) analyzed both 3 V and 6 V data sets.

During Run 21, both WIMP-search and calibration data were collected. Electron recoil response was recorded using a cobalt-60 source placed just inside the outer lead shielding, with three gamma-ray source calibrations performed at each bias (Akerib et al. 2011). Short calibration exposures were taken before data collection, followed by longer exposures during and after the WIMP-search runs (Akerib et al. 2011). These calibration data enabled event-by-event phonon signal corrections for position- and energy-dependent responses (Akerib et al. 2011). The modular phonon sensors provided xy-position (plane of the top and bottom detector surfaces) information that was used to correct position-dependent pulse-shape variations in the optimal-filter reconstruction, thereby improving phonon energy resolution (Akerib et al. 2011).

The detectors were exposed to a californium-252 neutron source with a section of their polyethylene shielding removed to calibrate their nuclear recoil response (Akerib et al. 2011). The authors performed these calibrations at the beginning and end of each WIMP search for each bias setting (Akerib et al. 2011). The resulting ionization yield distributions were used to define the expected nuclear recoil signal region and to estimate detection efficiencies (Akerib et al. 2011).

Although hardware thresholds occasionally triggered on electronic noise, the trigger rate was insufficient to characterize subthreshold noise distributions or near-threshold efficiency estimates (Akerib et al. 2011). To study these effects, data without the trigger bias (non-triggered data) were collected during the run using randomly generated software triggers, comprising roughly 5% of all events and resulting in a periodic sampling of electronic noise (Akerib et al. 2011). These samples, dominated by noise fluctuations, were useful for monitoring daily noise but inadequate for efficiency estimates due to the scarcity of near-threshold events (Akerib et al. 2011). In addition, a timing artifact in the random trigger produced unrepresentative traces, resulting in

non-Gaussian noise tails in the electronic noise distribution of each detector (Akerib et al. 2011). Scaling such data to WIMP search exposures resulted in grossly overestimated noise leakage, so non-triggered data were excluded from resolution and efficiency characterization (Akerib et al. 2011).

A separate dataset, free from trigger bias, was obtained during standard triggered readout (Akerib et al. 2011). While most triggers corresponded to energy depositions in one detector, the authors recorded and examined traces from all six detectors (Akerib et al. 2011). Data from the non-triggered detectors (called other-detector triggers) yielded an unbiased view of each detector's subthreshold noise distribution (Akerib et al. 2011). Non-Gaussian tails could be avoided if the noise sample excluded cases where the triggering detector was an adjacent detector (Akerib et al. 2011). When the trigger came from a detector directly above or below the detector under consideration, small multi-detector energy depositions were often present (Akerib et al. 2011). These events extended to recoil energies at and beyond the trigger thresholds, allowing examination of detector response near the thresholds (Akerib et al. 2011). In some of these events, the recorded energies were above the detector's trigger threshold, meaning they were not truly non-triggered events but rather events with delayed triggers that occurred during a readout initiated by another detector (Akerib et al. 2011). For each detector and event, these authors kept an electronic record of additional trigger signals relative to the main logical-pulse-based triggers, and this information formed the basis for their hardware trigger efficiency estimates (Akerib et al. 2011).

To qualify as potential WIMP events, signals had to surpass both software and hardware thresholds in phonon energy(Akerib et al. 2011). These authors determined detection efficiency in terms of reconstructed phonon energy using the other-detector triggers. Because the triggering energy and the corresponding offline reconstructed energy differed slightly, the hardware threshold efficiencies did not appear as simple step functions when plotted against the reconstructed values (Akerib et al. 2011). To characterize these efficiencies, the authors examined the reconstructed phonon energy within the 50-microsecond window following every other-detector trigger. The reconstructed phonon energies typically followed Gaussian distributions centered at zero, representing electronic noise ("noise cores") (Akerib et al. 2011). Occasionally, additional nonzero energy values appeared in reconstruction, producing a roughly uniform tail extending to higher energies (Akerib et al. 2011). The authors recorded logical pulses from the discriminator threshold in the post-trigger history of other-detector triggers. By selecting events with such logical pulses, the authors obtained the reconstructed phonon-energy distribution of successful triggers. Dividing this by the distribution of all triggers (both successful and unsuccessful) provided the hardware threshold efficiency, shown for a representative Ge detector in Figure 2 of Akerib et al (2011).

Akerib et al. (2011) set a software threshold on reconstructed phonon energy, assuming an approximately Gaussian distribution. Each day, a Gaussian fit was performed on the noise core of randomly triggered events (Akerib et al. 2011). The data from these randomly triggered events were recorded by the six

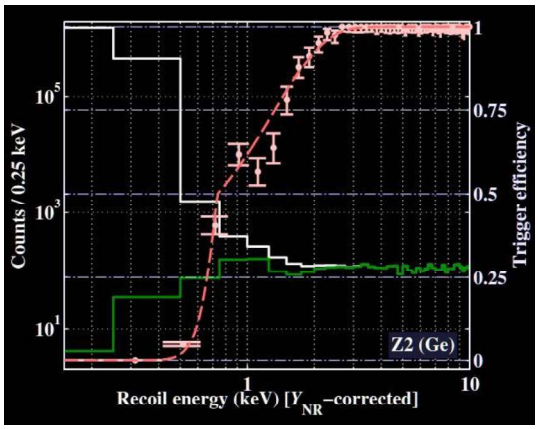

Figure 2: Hardware threshold efficiency estimate for one Ge detector (Z2 3V data) in Y_{NR}-corrected energy. The green solid line represents the recoil energy distribution that corresponds to the post-trigger history. The white solid line represents the reconstructed energy distribution. The efficiency is calculated by dividing the green solid line by the white solid line (Figure 2 from Akerib et al. 2011).

ZIP sensors, labeled Z1-Z6 (Akerib et al. 2011). Candidate WIMP events were required to surpass a threshold of 6σ above the average of that day (Akerib et al. 2011). For the Z2 and Z3 detector 3V data, these thresholds were highly stable, fluctuating by less than 5% each day, whereas the Z1 thresholds were much less stable, with fluctuations up to 50% (Akerib et al. 2011). Software threshold efficiencies were determined using the same procedure as for the hardware trigger efficiencies; the software efficiencies showed smooth variations due to daily changes in the Gaussian fit parameters and did not resemble idealized step functions (Akerib et al. 2011). The average hardware and software thresholds are summarized in Table II of Akerib et al.(2011). The Z1 detector was excluded from Table II (Akerib et al. 2011) because it failed as a low-threshold detector. The pulse height from the Z1 data showed an overly strong dependence on xy-position that the gamma-ray correction could not completely compensate for (Akerib et al. 2011). As a result, the corrected resolution and corresponding 6σ thresholds for Z1 were three to four times higher than the resolutions of the other Ge detectors (Akerib et al. 2011). This raised the effective analysis threshold and caused ionization-yield discrimination to fail at higher energies (Akerib et al. 2011). Because sensitivity to low-mass WIMPs depends strongly on the detection threshold, the Z1 data added little to the overall analysis while disproportionately increasing background leakage into the signal region (Akerib et al. 2011). For this reason, Z1 was used solely as a veto for multiple-scattering events (Akerib et al. 2011).

Final threshold efficiencies for each detector were calculated by taking the product of their hardware and software efficiencies (Akerib et al. 2011). These

	Hardware (keV)		Software (keV)	
Detector	3V	6V	3V	6V
Z2	$0.74^{+0.07}_{-0.02}$	$0.67^{+0.09}_{-0.02}$	0.63 ± 0.01	0.58 ± 0.02
Z3	$1.13^{+0.07}_{-0.04}$	$1.12^{+0.07}_{-0.05}$	0.82 ± 0.02	0.72 ± 0.03
Z4	$1.77^{+0.12}_{-0.10}$	$1.71^{+0.13}_{-0.10}$	$1.62^{+0.10}_{-0.09}$	$1.52^{+0.09}_{-0.08}$
Z5	$1.00^{+0.05}_{0.04}$	$0.91^{+0.06}_{0.05}$	0.73 ± 0.02	$0.63^{+0.04}_{0.03}$
Z6	$1.53^{+0.08}_{-0.06}$	$1.55^{+0.09}_{-0.06}$	$1.39^{+0.06}_{-0.05}$	1.34 ± 0.05

Table 2: Average hardware and software phonon energy thresholds with 1σ errors in Y_{NR}-corrected energies for 3V and 6V data (Table II from Akerib et al. 2011).

software efficiencies, as a function of reconstructed energy, resembled step functions, reaching 100% at lower energies than the hardware efficiencies. The combined curves differed from the hardware efficiency curves only at very low energies (Akerib et al. 2011). Figure 3 from Akerib et al. (2011) shows these combined efficiencies for representative Ge and Si detectors. For nearly all of the six ZIP detectors, the 50% efficiency points matched the hardware thresholds listed in Table II of Akerib et al. (2011).

Candidate WIMP events were required to satisfy strict data-quality criteria. Any event with incomplete or inconsistent data records was discarded (Akerib et al. 2011). The authors rejected traces that were excessively noisy or showed multiple pulses ("pileup") based on optimal-filter fits to the ionization signals, and events that had anomalously high χ^2 values were also excluded (Akerib et al. 2011). Additional suppression of noise and pileup was achieved by requiring the prepulse baseline variances of both ionization and phonon traces to remain within 5σ of the mean behavior observed in randomly triggered events (Akerib et al. 2011). These quality cuts were energy-independent and yielded an overall WIMP detection efficiency of about 99%, consistent across detector types and for both the 3V and 6V data sets (Akerib et al. 2011).

The standard data-quality cuts would have sufficed if this analysis were restricted to recoil energies above 2 keV (Akerib et al. 2011). However, during the 6V run, the Z2 Ge detector experienced intermittent high-trigger-rate periods ("event bursts") of sub-2 keV pulses, often following cryogenic or servicing periods when detector temperatures were elevated (Akerib et al. 2011). These bursts were not attributed to genuine recoils and were treated as poor-quality data (Akerib et al. 2011). Inspection of the phonon traces showed elevated prepulse baselines, allowing these events to slip past the usual quality cuts (Akerib et al. 2011). To address this, the authors introduced new parameters for discrimination based on mean prepulse baselines, effectively removing the bursts. Applying this cut reduced the usable exposure of Z2 to 20 days and lowered the number of candidate events by roughly a factor of 30 (Akerib et al. 2011).

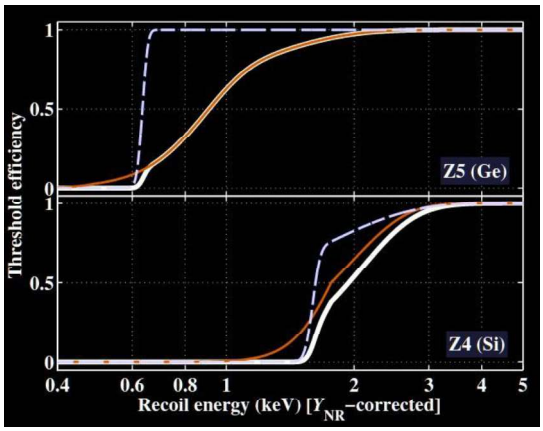

Figure 3: The orange solid line represents the estimated hardware phonon energy threshold efficiencies, and the blue dashed line represents the estimated software phonon energy threshold efficiencies. The white solid line is the product of the hardware and software efficiencies, and is defined as the combined threshold efficiency. The figure displays these data for representative Ge (Z5) and Si (Z4) detectors (Figure 3 from Akerib et al. 2011).

Although confined to one detector and under half of the WIMP-search dataset, the burst events would have dominated the signal region background without this data cut (Akerib et al. 2011).

As noted above, Akerib et al. (2011) used a fiducial-volume cut determined by ionization signals to exclude recoils occurring near detector edges. The outer electrode (q-outer) signal was required to remain consistent with noise while the inner electrode (q-inner) signal provided the ionization energy estimate (Akerib et al. 2011). The q-outer signal was slightly increased by a contribution proportional to the q-inner signal amplitude, so the q-outer noise was parameterized as a function of q-inner (Akerib et al. 2011). This defined 2σ q-outer "noise bands" that retained 95.5% of fiducial-volume events (Akerib et al. 2011). Additional signals induced in the electrodes by the phonon sensors of neighboring detectors initially led to viable events being lost (Akerib et al. 2011). To recover them, the authors modified the cut to reject only events above the upper noise-band limit, raising the expected acceptance to 97.7% (Akerib et al. 2011). Combining this acceptance with the surface area of the inner electrode yielded a fiducial volume of about 83% of the detector mass (Akerib et al. 2011). Calibration using californium-252 nuclear recoils between 4-100 keV confirmed the expected efficiency, which ranged from roughly 81% (Z4, 3V data) to roughly 83% (Z2, 3V data) (Akerib et al. 2011).

Across the measurable recoil-energy range, the fiducial-volume cut efficiency shows only a weak energy dependence, generally declining with increasing energy (Akerib et al. 2011). The strongest trend appears in the Z4 6V data, where efficiency decreases smoothly from about 84% at 4 keV to roughly 79%

at 100 keV (Akerib et al. 2011). Below 4 keV, it becomes difficult to distinguish electronic noise from real nuclear recoil signals from q-inner and q-outer. This prevents the cut from reliably separating fiducial-volume events from edge events (Akerib et al. 2011). Because acceptance should increase at lower energies, the fiducial-volume cut efficiency is expected to grow rapidly as the recoil energy approaches 0 keV (Akerib et al. 2011). However, since this low-energy behavior was not directly measured, Akerib et al. (2011) decided to linearly extrapolate the efficiency below 4 keV.

Because true but infrequent WIMP interactions are expected to deposit energy in only one detector, unlike background processes, Akerib et al. (2011) required candidate events to have deposited significant energy in only one detector (single scatter criterion). Since the phonon channel is the most sensitive probe of particle interactions, this cut was defined using the 6σ software thresholds described earlier and in Table II of Akerib et al. (2011). Given the experimental trigger rate of $\lesssim$1 Hz, the chance of two unrelated pulses overlapping within the $\lesssim$2 ms digitization window is negligible (Akerib et al. 2011). As a result, the single scatter cut had an efficiency of almost 100%, with the small near-threshold effects already incorporated into the software threshold efficiency shown in Figure 3 of Akerib et al. (2011).

Candidate WIMP events were required to occur without muon coincidence alerts (Akerib et al. 2011). For each ZIP-triggered event, these alerts were recorded at an average rate of roughly 5 kHz, much higher than the true incident muon rate (Akerib et al. 2011). Because the scintillator counters had irregular geometries and regions of suboptimal light collection, the photomultiplier tubes were operated at a high gain to allow efficient muon detection (Akerib et al. 2011). This made regions with strong light collection, especially near the photomultiplier tubes, partially sensitive to environmental gamma rays (Akerib et al. 2011). A high fraction of ZIP interactions coincident with these gamma-induced detections was rejected, yielding a muon-tagging efficiency above 99.9% (Akerib et al. 2011). An event was vetoed if any coincident detection occurred within 50-80 microseconds before the triggering phonon pulse, a window chosen to account for the slower rise of phonon signals compared to the nearly instantaneous coincidence detection (Akerib et al. 2011). For recoil energies under roughly 3.5 keV, the longer delay ensured that small phonon pulses had time to cross the hardware threshold (Akerib et al. 2011). Inevitably, some WIMP detection efficiency was lost due to gamma ray-induced coincident detections, leaving net per-detector efficiencies of roughly 67% to 78% (Akerib et al. 2011).

The final event selection requires that the measured ionization yield be consistent with a nuclear recoil, which suppresses the dominant background from electron recoils (Akerib et al. 2011). These authors collected calibration data through short exposures to neutrons from a californium-252 source, providing adequate nuclear-recoil data to characterize the ionization yield response for each detector as a function of recoil energy. The ionization yield distribution for nuclear recoils is roughly a Gaussian function down to 2 keV when binned in recoil energy (Akerib et al. 2011). The average yield and corresponding Gaussian width for these data were determined in each bin using maximum-

likelihood fits. Nuclear-recoil ionization yield Y in semiconductor crystals was modeled as,

$$Y = a \cdot E_{recoil}^{b} \tag{5}$$

where a and b are fit parameters, and E_{recoil} is the recoil energy (Akerib et al. 2011). This model was fit to a set of mean values, while the squared widths were modeled with an inverse squared form:

$$\sigma_Y^2 = d^2 + c^2/E_{recoil}^{\ 2} \tag{6}$$

where σ_Y is the width of the ionization yield Gaussian function, and c and d are fit parameters. (Akerib et al. 2011). These authors then used the fitted trends to define nuclear-recoil acceptance bands. An event is accepted as a nuclear recoil if the ionization yield is within 2σ of the average. Figure 4 from Akerib et al. (2011) illustrates an example acceptance band and mean-yield fit relative to the underlying californium-252 calibration data. The band selection efficiency for each detector is described well by an energy-independent value ranging from roughly 93% (for the Z6 3 V data) to 96% (for the Z3 and Z4 3 V data) (Akerib et al. 2011). Figure 5 of Akerib et al. (2011) shows the combined cut efficiency as a function of recoil energy for the chosen detector. The discontinuity between 3 keV and 4 keV reflects the adoption of a wider muon-veto window at low energies (Akerib et al. 2011). Also plotted in this figure is the 90% confidence-level lower bound on the efficiency, defined as 1.28σ below the mean, which quantifies the statistical precision of the efficiency estimate (Akerib et al. 2011).

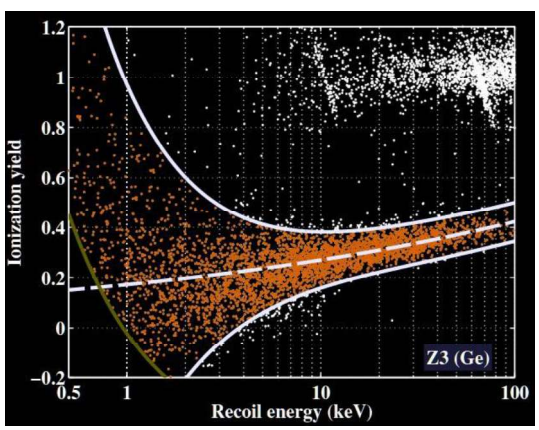

Figure 4: Ionization yield vs recoil energy for the californium-252 calirbration data for a representative Ge detector (Z3 6V data). The blue dashed line represents the average ionization yield, the white solid lines represent the boundaries of the 2σ recoil band, and the yellow solid line represents the software phonon threshold (Figure 4 from Akerib et al. 2011).

The six ZIP detectors had two distinct energy scales (Akerib et al. 2011). One scale was defined for electron recoils and the other was defined for nuclear

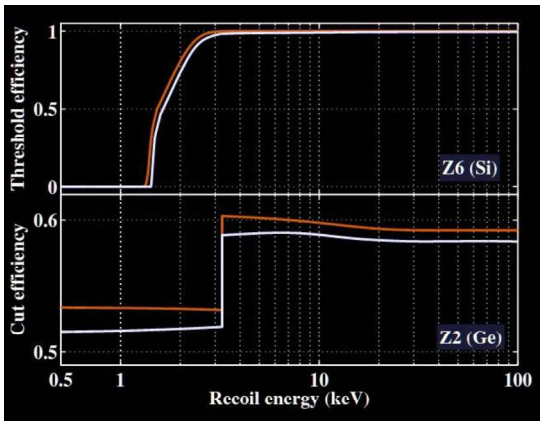

Figure 5: Detection efficiencies of hardware and software phonon thresholds and analysis cuts for Si Z6 3V data (top panel) and Ge Z2 6V data (bottom panel) for Run 21 of the CDMS shallow-site experiment (Figure 5 from Akerib et al. 2011).

recoils. The authors calibrated the electron-recoil scale at the start of Run 21 using 662 keV gamma-rays from a cesium-137 source (Akerib et al. 2011).

The calibration spectrum of a representative Ge detector is shown in Figure 6 (Figure 6 from Akerib et al. 2011). The electron-recoil energy scale of each Ge detector was calibrated using three characteristic emission peaks spanning the analysis energy range (Akerib et al. 2011). Two of these peaks arise from unstable isotopes: germanium-68, produced by cosmic rays, and germanium-71, produced by thermal neutron capture. These sources produce 1.3 keV and 10.4 keV decay products (x-rays or electrons), yielding the first two peaks in Figure 6. A third calibration peak originates from germanium-73m, which sequentially produces 53.4 keV and 13.3 keV photons, yielding the 66.7 keV peak in Figure 6. To enhance resolution, the spectrum in Figure 6 is plotted as a function of Y_{ER}-corrected recoil energy, and an analogous spectrum with the same features can be obtained from the ionization channel (Akerib et al. 2011). The Ge decay emission lines were used to measure the energy resolutions of both the phonon and ionization channels across the analysis range. For electron recoils, the fractional energy resolutions spanned 8.8% for the Z5 6 V phonon channel at 1 keV to 26.0% for the Z3 3 V ionization channel at 1 keV (Akerib et al. 2011). The fractional energy resolutions spanned 2.0% for the Z3 and Z5 eV ionization channels at 100 keV to 5.4% for the Z2 3V phonon channel at 100 keV (Akerib et al. 2011). The silicon detectors lack isotope-decay features analogous to those in the Ge detectors. Consequently, the Si detector resolutions could only be assessed at zero energy, where they are determined entirely by noise (Akerib et al. 2011).

In a low-threshold analysis, it is necessary to account for nonzero energy resolutions near the detection threshold. For all the ZIP detectors in Akerib

et al. (2011), electronic noise dominated the resolution in the near-threshold regime. Accordingly, when calculating WIMP exclusion limits, these authors applied only the zero-energy resolution where required. This approach yielded exclusion limits that were more conservative than those obtained by including the energy-dependent terms, while also enabling an equal treatment of both Si and Ge detectors (Akerib et al. 2011).

To calculate the WIMP exclusion limits, Akerib et al. (2011) used both the Q-corrected and Y_{NR}-corrected resolutions. The Y_{NR}-corrected resolution accounts for subthreshold pulse triggers due to variations in phonon-noise and therefore excludes contributions from the ionization channel (Akerib et al. 2011). The Q-corrected resolution reflects the resolution intrinsic to the overall analysis-cut efficiency, incorporating both ionization and phonon noise (Akerib et al. 2011). These authors obtained measurements of these resolutions at 0 keV from other-detector-triggered data, and the results for the viable low-threshold detectors are summarized in Akerib et al. Table III (2011).

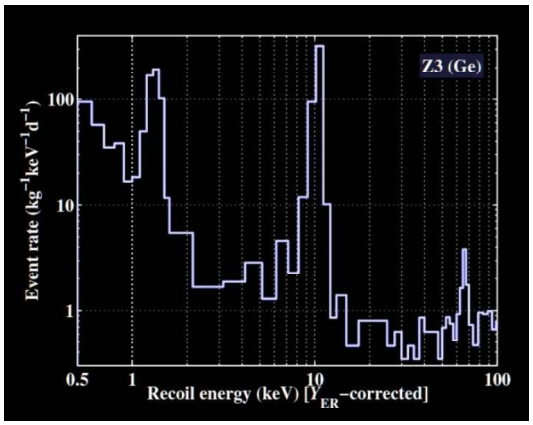

Figure 6: $Y_E R$-corrected recoil energy vs. event rate for a representative Ge detector (Z3 6V) during WIMP-search data collection in Run 21 of the CDMS shallow-site experiment (Figure 6 from Akerib et al. 2011).

The nuclear-recoil energy scale is critical for WIMP searches, while electron-recoil calibration is mainly useful for evaluating detector performance. Akerib et al. (2011) determined this scale by comparing californium-252 calibration data with a GEANT3 simulation of the source and detector geometry. To increase statistical precision, recoil events were combined into mean Ge and Si spectra (Akerib et al. 2011). As shown in Figure 7 (Figure 7 of Akerib et al. 2011), the Si spectra are well fit by a single decaying exponential, and the Ge spectra by two decaying exponentials. Comparing the exponential decay fit constants provides an absolute-rate-discrepancy-independent test of the recoil-energy scale (Akerib et al. 2011). For the Si detectors, these authors observed a 16±3% energy-scale discrepancy, which was corrected by shifting the Si scale upwards by 16%, yielding a more conservative upper limit (Akerib et al. 2011). The

	Y_{NR}-corrected (eV)		Q-corrected (eV)	
Detector	3V	6V	3V	6V
Z2	95	88	223	387
Z3	126	114	282	477
Z4	208	196	398	538
Z5	113	102	258	450
Z6	185	179	434	678

Table 3: Zero energy recoil-energy resolution for viable low-threshold detectors for 3V and 6V data collection runs in Run 21 of the CDMS shallow-site experiment (Table III from Akerib et al. 2011)

Ge detectors show smaller deviations, with overestimation at low energies and underestimation at high energies, relative to the Monte Carlo spectra (Akerib et al. 2011). These authors note that this deviation was not corrected, as electron-recoil calibration is more reliable and the deviation is small. This choice results in a more conservative exclusion limit for low-mass WIMPs (Akerib et al. 2011).

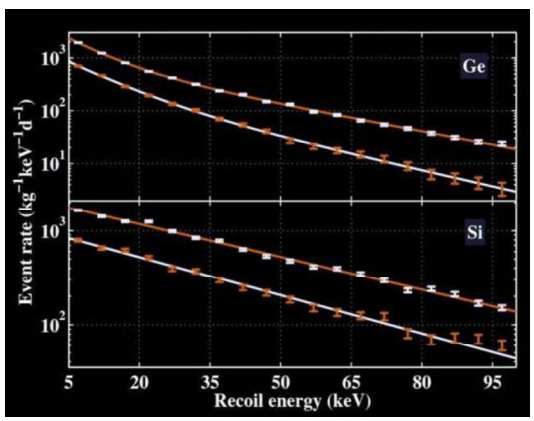

Figure 7: Monte Carlo simulated spectra compared to experimental spectra for Ge and Si detectors for Run 21 of the CDMS shallow-site experiment (Figure 7 from Akerib et al. 2011).

The potential impact of ion channeling on direct-detection energy scales has been widely discussed since Sekiya et al. (2003). Early models for NaI scintillation detectors, as detailed in Drobyshevski (2008) and Bernabei et al. (2008), showed that a non-negligble fraction of recoiling nuclei could channel between detector crystal planes, reducing interaction strength with atomic cores and depositing a larger fraction of their energy as ionization (Akerib et al. 2011).

In such cases, the full recoil energy from a WIMP interaction could appear as an electron-recoil signal, effectively lowering the nuclear-recoil threshold in NaI scintillation detectors (Akerib et al. 2011). For the Ge and Si detectors in Akerib et al. (2011), the higher ionization yield due to channeling would shift nuclear recoil signals toward the electron-recoil signal band, thereby reducing nuclear-recoil acceptance. If channeling is especially significant at low recoil energies, as suggested in Bernabei et al. (2008), the sensitivity to low-mass WIMPs could be substantially degraded. Akerib et al. (2011) therefore calculated an upper limit to the fraction of near-threshold channel events using the californium-252 calibration data. For recoil energies between 2-6 keV, fewer than 4% of events had ionization yields above the nuclear-recoil band (Akerib et al. 2011). This apparent excess over the rate of background electron recoils was attributed to secondary gamma rays: 2.2 MeV photons from neutron capture on hydrogen in the polyethylene shielding, and roughly 1 MeV photons emitted directly by the californium-252 source due to fission (Akerib et al. 2011). Therefore, they found no evidence for a channeling effect large enough to significantly impact efficiency estimates and thus neglected ion channeling. This is consistent with the early models employed by Bozorgnia et al. (2010) showing a negligible effect for low-energy recoils in cryogenic Ge and Si detectors (Akerib et al. 2011).

3 Results

3.1 EDELWEISS

Figure 3 of Benoit et al. (2002) presents the recorded data within the fiducial volume of GGA1. The ionization-to-recoil energy ratio is plotted against recoil energy, showing events triggered exclusively by GGA1 with ionization energies above 3.5 keV. The 99.9% photon acceptance band is derived from the mean heat and ionization resolutions under the assumption of Gaussian dispersion. A small cluster of events near $Q \sim 0.5$, linked to low-energy beta and gamma interactions—previously prominent in Di Stefano et al. (2000) and less so in Benoit et al. (2001)—is represented here by only four to five events (Benoit et al. 2002). Greater exposure is required for proper interpretation of this cluster. A similar limitation applies to the three events with energies between 119 keV and 182 keV observed below $Q = 0.7$ (Benoit et al. 2002).

The 119 keV event observed with $Q = 0.3$ lies at -1.646σ from the center of the nuclear recoil band (Benoit et al. 2002). Due to uncertainties in σ, the authors conservatively classify this as a recoil included in the acceptance region for masses above 10 TeV/c^2. For lower WIMP masses, no events fall within the nuclear recoil band. The lack of accepted recoils is therefore used to set 90% confidence level upper limits on the WIMP–nucleon scattering cross section for $M_W < 10$ TeV/c^2, following the procedure outlined in Lewin and Smith (1996) with standard halo and nuclear models (Benoit et al. 2002).

Figure 4 of Benoit et al. (2002) displays the exclusion limit as a function of WIMP mass. This result is compared with the earlier measurement obtained

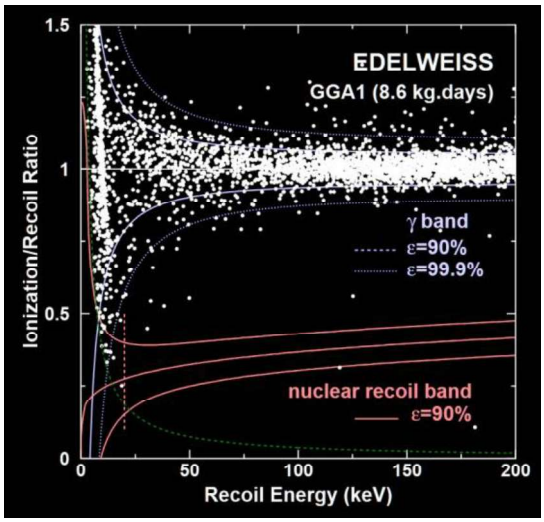

Figure 8: Distribution of ratios of ionization signals to recoil energies as a function of recoil energy from GGA1 fiducial volume data with 8.6 kg ·d. The dotted line represents the 99.9% efficiency region for photons (Figure 3 of Benoit et al. 2002).

using the GeAl6 detector, which had an effective exposure of 4.3 kg·d [**?**]. The combined limit from both datasets, corresponding to a total effective exposure of 11.7 kg·d, is also presented. For reference, the 3σ contour associated with the annual modulation signal reported by DAMA NaI1-4 [**?**] is shown, with the black circle indicating the central value at M_W = 52 GeV/c^2 and $\sigma_n = 7.2 \times 10^{-6}$ pb.

3.2 CDMS

After Akerib et al. (2011) applied their analysis cuts and phonon software thresholds, they detected a significant residual event rate in the low-threshold signal region. These authors focused on the 0.5-100 keV energy region for Ge detectors and the 1-100 keV region for Si detectors, leaving 1080 candidate events for Ge and 970 candidate events for Si. The raw live days and corresponding event counts for each detector and bias voltage are summarized in Table IV of Akerib et al. (2011), while their Figure 8 shows the combined Ge and Si spectra with analysis cut and hardware/software threshold corrected event rates. Because both the spectra and the analysis cut efficiencies are defined in terms of Q-corrected recoil energy, the threshold efficiencies were first converted from YNR-corrected to Q-corrected recoil energy prior to applying the threshold corrections (Akerib et al. 2011).

While the recoil spectra display shapes consistent with those from WIMP-induced interactions, the events observed within the signal region are probably from background processes (Akerib et al. 2011). These include electron recoils,

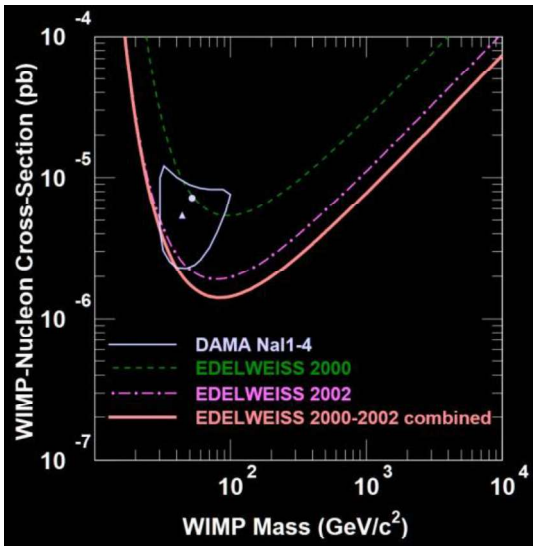

Figure 9: Exclusion limits for WIMP-nucleon spin-independent cross-section as a function of WIMP mass with 11.7 kg ·d of exposure. The dashed curve represents previous EDELWEISS data from Benoit et al. (2001). The dash-dotted line represents data from Benoit et al. (2002) (Figure 4 of Benoit et al. 2002).

zero-ionization events, carbon contamination largely localized to detector Z6, and nuclear recoils generated by cosmogenic neutrons. Instead of subtracting these contributions, Akerib et al. (2011) classified all such events as potential candidates when establishing upper bounds on a WIMP signal, thereby adopting the most conservative treatment of the dataset.

Distinct background populations are clear in ionization yield–recoil energy plots, with representative signal regions and candidate events for Ge and Si detectors illustrated in Fig. 9 (Akerib et al. 2011). The Ge detectors show the most prominent background, manifesting as a 1.3 keV feature in the 1–3 keV recoil-energy range that contributes significantly to the candidate event count (Akerib et al. 2011). This feature originates from internal electron-captured decays of germanium-68 and germanium-71, whose associated x-rays and electrons yield unit ionization. At these low energies the limited signal-to-noise ratio in both phonon and ionization channels causes electronic noise to produce an extended tail of electron recoils with reduced ionization yield (Akerib et al. 2011). The numerator and denominator of the ionization yield definition exhibit anticorrelation, causing the 1.3 keV feature to appear tilted relative to the recoil-energy axis. Roughly 20% of Ge candidates originated from electron recoils leaking from the 1.3 keV line in the 3 V WIMP search, while this process caused between one-third (Z3) and one-half (Z2, Z5) of Ge candidates in the 6 V search. The larger fraction of events observed in the 6 V data set is due

	3V Exposure		6V Exposure	
Detector	Live Days	Candidates	Live Days	Candidates
Z2	66.12	159	20.16	67
Z3	66.12	129	51.66	349
Z4	66.12	130	51.66	125
Z5	66.12	174	51.66	202
Z6	66.12	401	51.66	314

Table 4: Number of raw live days and number of WIMP candidate events corresponding to each detector following analysis cut and threshold corrections for Run 21 of the CDMS shallow-site experiment (Table IV from Akerib et al. 2011).

to californium-252 neutron calibrations performed prior to that run, which enhanced the germanium-71 population through thermal neutron capture (Akerib et al. 2011).

Both Ge and Si devices exhibit a class of events characterized by near-zero ionization yield, but this effect is more pronounced in the Si detectors (Akerib et al. 2011). Closer inspection shows that these "zero-charge" events produce ionization signals indistinguishable from electronic noise, despite otherwise resembling ordinary recoils with energies extending up to 100 keV. In the Si detector shown in Fig. 9 of Akerib et al. 2011, the zero-charge event distribution spans the full analysis energy range and extends into the signal region at $\lesssim 7$ keV, constituting the majority of candidate events in this detector. An explanation for this phenomenon is that recoils occurred where the electric field lines failed to span the thickness of the detectors, near the side edges of the detectors (Akerib et al. 2011). If ionization is generated along field lines that terminate on a detector edge rather than an electrode, it is not drifted properly, resulting in zero signal in the inner or outer ionization channels. For recoil energies exceeding 10 keV, the xy-positions of zero-charge events can be reconstructed and cluster near detector edges. Previous CDMS analyses suppressed the zero-charge background by requiring a minimum ionization threshold for WIMP candidates, analogous to the phonon software thresholds. However, Akerib et al. (2011) chose to discard this requirement, as imposing such a threshold would significantly reduce detection efficiency for recoil energies below 5 keV.

Akerib et al. (2011) attributed the elevated event rate observed in detector Z6 to surface contamination by carbon-14. During testing at a CDMS facility before Run 21, detector Z6 was operated close to a detector that had been exposed to a carbon-14 calibration source with poor shielding, resulting in a small contamination of carbon-14 on one surface of Z6. To mitigate cross-contamination, Akerib et al. (2011) positioned Z6 at the bottom of the detector tower with the affected surface oriented away from neighboring detectors. Carbon-14 beta decays yield electrons with a maximum energy of about 156 keV and an average energy of about 50 keV. The radiation in this energy range interacts only within

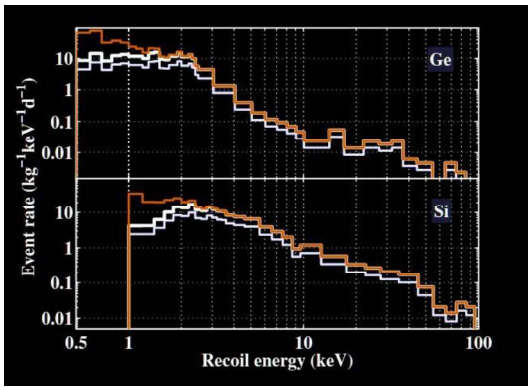

Figure 10: Combined Ge (top) and Si (bottom) WIMP candidate event rates versus recoil energy for Run 21 of the CDMS shallow-site experiment. The figure compares raw event rate data (blue solid lines) and efficiency corrected event rates. The corrected rates are corrected by exposure-weighted detector-averaged analysis cut efficiencies (white solid lines), and then by detector averaged hardware/software thresholds (orange solid line) (Figure 8 from Akerib et al. 2011).

the roughly 10 μm surface dead layer of the ZIP detectors (Akerib et al. 2011). Consequently, these events exhibit suppressed ionization yield and occupy the gap between electron- and nuclear-recoil bands when plotted as in Fig. 9 of Akerib et al. (2011), with a considerable fraction leaking into the signal region. While surface events can be efficiently rejected at recoil energies above 10 keV using combined ionization and phonon pulse-timing parameters, such discrimination cannot be maintained without compromising the near-threshold WIMP detection efficiency (Akerib et al. 2011).

Electron recoil leakage into the nuclear-recoil band constitutes a background component of candidate events in all detectors, typically arising from Compton scattering of photons (Akerib et al. 2011). The ionization-yield discrimination degrades as recoil energy decreases, leading to significant overlap of the electron- and nuclear-recoil bands at the "crossover energy." This crossover varies between approximately 3 keV (Z5, 3 V bias) and 7 keV (Z6, 6 V bias). Operation at higher bias voltage produces additional phonons, which in turn degrade both recoil-energy resolution and yield-based discrimination (Akerib et al. 2011). As a result, ZIP detectors achieve superior low-threshold performance at the lower 3 V bias. Calibration data scaled from a cobalt-60 source indicate that only a small number of Compton events leak into the nuclear-recoil band for energies above the crossover energy (Akerib et al. 2011). For recoil energies below this threshold, however, no reliable method has been established to quantify the leakage contribution. Akerib et al. (2011) estimated that 10–20% of the WIMP candidates were electron recoils originating from Compton scatters.

Neutron backgrounds arising from the limited overburden of SUF are the most likely cause of the highest-energy signal events (Akerib et al. 2011).

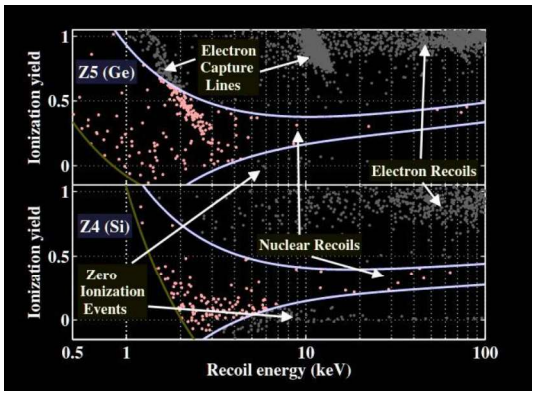

Figure 11: Figure 9 from Akerib et al. 2011 shows ionization yield versus recoil energy for unshielded single scatters that pass the quality and fiducial-volume cuts for the Ge Z5 6V data (top) and Si Z4 3V data (bottom) WIMP searches. The signal regions for each detector are outlined by the nuclear-recoil band (blue solid lines) and phonon energy software threshold (yellow solid lines). All Z5 6V and Z4 3V WIMP candidate events (red dots) can be seen in these signal regions.

Muons, together with hadronic showers they initiate, occasionally fragment nuclei in the surrounding rock, releasing high-energy neutrons capable of penetrating the shielding and producing recoil signals above threshold (Akerib et al. 2011). The expected spectrum of unshielded neutron interactions in the Ge detectors was simulated as shown by the dashed line in Figure 2 of Akerib et al. (2003). Without efficiency corrections, the rate peaks at approximately 0.1 events/kg/keV/day at the 0.5 keV threshold and decreases to about 0.003 events/kg/keV/day at 100 keV (Akerib et al. 2011). Incorporating the average detection efficiencies of the Ge detectors and scaling to the total exposure of ~71 kg·days yields ~66 expected neutron events among the 1080 Ge candidates (Akerib et al. 2011).

A precise determination of the relative contributions from individual background sources to the total candidate population is challenging (Akerib et al. 2011). The majority of candidate events occur at low recoil energies, where the available experimental variables lack sufficient discriminatory power, particularly in distinguishing zero-charge events from Compton electron-recoil leakage. Table V of Akerib et al. (2011) provides an approximate breakdown of the candidate event percentages attributable to each background source for the combined Ge and Si detector ensembles. An additional "other" category is included to represent the fraction of events not accounted for by the identified backgrounds. These residual events may originate from unidentified background processes or from WIMP interactions, but are likely due to uncertainties corresponding to known background contributions (Akerib et al. 2011).

Due to substantial uncertainties in background estimation, the authors were

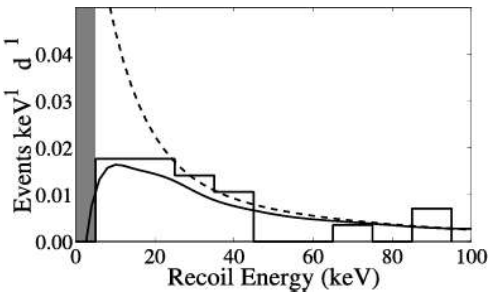

Figure 12: The dashed curve depicting the shape of expected recoil-energy spectra due to incident neutrons (Figure 2 from Akerib et al. 2003).

	Ge (%)	Si (%)
Electron Capture 1.3 keV Line	32	0
Zero-Charge Events	30–40	30–40
^{14}C Contamination Betas	0	40
Compton Photon Electron Recoils	10–20	10–20
Cosmogenically-Induced Neutrons	6	2
Other	2–22	0–18

Table 5: Table V from Akerib et al. 2011 shows percentages of the 1080 Ge and 970 Si candidate events attributed to each background contribution.

unable to subtract the background from the data. Therefore, no evidence for a WIMP signal was established; instead, the authors were able to derive conservative limits because the candidate events may include WIMP interactions. Using this, Akerib et al. (2011) constrained the spin-independent WIMP-nucleon elastic scattering cross section using observed event rates. The principal result presented by the authors combines Ge and Si data from all 2050 candidate events, while the secondary result is based solely on the Si dataset.

Exclusion limits are derived by comparing observed event rates with a hypothetical WIMP model (Akerib et al. 2011). For consistency with other direct detection experiments, the authors adopt the standard halo model described by Lewin and Smith (1996), but instead use a normalized local WIMP density of 0.3 GeV/cm^3. Akerib et al. (2011) also use a WIMP characteristic velocity of 220 km/s and a mean Earth velocity of 232 km/s (Donato, Fornengo, and Scopel 1998), and they also use a galactic escape velocity of 544 km/s (Smith et al. 2007). These values are crucial in calculating exclusion limits (Akerib et al. 2011).

The authors calculated these exclusion limits using a form of the optimum interval method (Yellin 2002), adapted for high-statistics data (Yellin 2007). For each WIMP mass, the limit is governed by the energy interval in which observed

events are notably fewer than expected WIMP events (Akerib et al. 2011). The authors applied statistical penalty that accounts for the selection of the most constraining interval, setting a 90% confidence-level upper bound. This approach is particularly effective in rejecting backgrounds with distributions distinct from the anticipated signal.

To obtain a single exclusion limit, Akerib et al. (2011) combined the Ge and Si detector data. The authors used the traditional CDMS method that combines detector data into a single averaged data set, in which detector masses and efficiencies are weighted by exposure. This method incorporates the full exposure and merges candidate event energies across detectors prior to constructing the energy intervals required by the optimum interval technique. Figure 10 (Figure 8 of Akerib et al. 2011) shows the averaged candidate event distribution used in this analysis. The averaging procedure is valid when detectors exhibit comparable sensitivity to WIMP interactions, as in earlier CDMS searches that were background-free or nearly so (Akerib et al. 2011).

Applying the averaging method to detectors with differing event rates causes high-rate detectors to obscure the sensitivity of low-rate detectors by populating critical intervals disproportionately (Akerib et al. 2011). To mitigate this, the authors adopted a "serialization" approach, in which energy intervals are defined separately for each detector to preserve sensitive intervals and concatenated in an arbitrarily predetermined order to avoid bias. In the analysis presented by the authors, 3V data precede 6V data, followed by ordering within the detector tower from top to bottom. When intervals do not span multiple detectors, the ordering has no effect on the result (Akerib et al. 2011). This method enables the optimum interval technique to include the most sensitive detector-specific intervals in calculating the exclusion limits, though the resulting limit reflects only a fraction of the total exposure. The data from each detector is heavily limited by the background, especially near threshold, so the authors elected to sacrifice the exposure for cleaner intervals, which was expected to improve limit strength at low masses. This proved effective for WIMP masses below 8 GeV/c^2. Moreover, serialization naturally accommodates combining detector types (Akerib et al. 2011).

Akerib et al. (2011) calculated expected WIMP rates separately for each detector and dataset (3V and 6V) to incorporate non-zero energy resolution. The authors calculated limits for 75 WIMP masses between 1–100 GeV/c^2. For each mass, the halo model provides the ideal recoil-energy spectrum, which the authors convolved with the YNR-corrected resolution (Table III of Akerib et al. 2011). Because software and hardware thresholds only depend on the phonon signal, only phonon noise was included when applying threshold efficiencies. The authors then further convolved the spectrum to account for ionization-channel noise, using the quadrature difference between YNR-corrected and Q-corrected resolutions. The threshold-reduced spectrum, expressed as a function of Q-corrected recoil energy, was multiplied by analysis cut efficiencies and scaled by detector mass and exposure (Tables I and Table IV of Akerib et al. 2011). These doubly convoluted, efficiency-reduced spectra yielded expected event distributions per Q-corrected keV, from which the authors constructed cumulative

probabilities for each detector that describes the likelihood that a WIMP interaction occurred within the candidate event energy interval of the detector. The resulting 10 (Ge and Si) or 4 (Si-only) probability distributions were serialized to compute optimum interval upper limits at each WIMP mass (Akerib et al. 2011). While non-zero resolution was carefully treated here, its impact would have been negligible in earlier CDMS analyses due to higher thresholds (Akerib et al. 2011).

Figure 13 (Figure 10 of Akerib et al. 2011) presents the calculated exclusion limits, which partially constrain parameter space associated with signals from DAMA/LIBRA (Savage et al. 2009, Bernabei et al. 2008, Bernabei et al. 2010) and CoGeNT (Aalseth et al. 2010). The DAMA/LIBRA region incorporates the aforementioned minor ion-channeling effect (Bozorgnia et al. 2010). The combined Ge and Si limit intersects the center of the 99% confidence-level region from a joint fit to DAMA/LIBRA and CoGeNT data carried out by Hooper et al. (2010), excluding new parameter space for WIMP masses between 3–4 GeV/c^2. Figure 14 (Figure 11 of Akerib et al. 2011) compares these limits with results from other experiments sensitive to low-mass WIMPs.

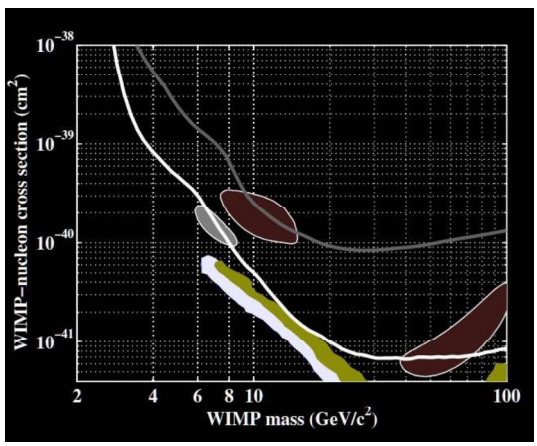

Figure 13: Comparison of 90% confidence level upper limits from combined (Ge and Si, black solid line) and Si detector only (gray solid line) data from Run 21 of the CDMS shallow-site experiment. This figure includes signal regions from DAMA/LIBRA and CoGeNT data. The two pink shaded regions represent the DAMA/LIBRA annular modulation signal and include ionization channeling effects. The gray shaded region is the 99% confidence level region from Hooper et al. (2010). The long yellow region are SUSY theory predictions for $\Omega_{WIMP} < \Omega_{CDM_{min}}$ (yellow region) and $\Omega_{WIMP} \geq \Omega_{CDM_{min}}$ (white region) (Figure 10 from Akerib et al. 2011)

Akerib et al. (2011) calculated a Ge-only limit and found it to be nearly identical to the combined Ge and Si limit, with differences imperceptible between Figure 13 and Figure 14. The serialization method sets exclusion limits

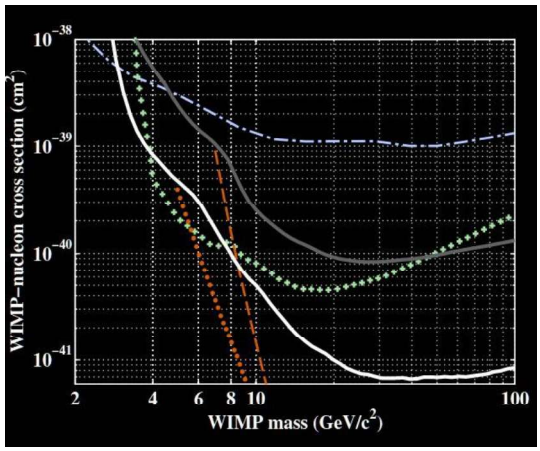

Figure 14: Comparison of 90% confidence level upper limits from combined (Ge and Si, white solid line) and Si detector only (gray solid line) data from Run 21 of the CDMS shallow-site experiment with data from the XENON100 experiment with constant (orange dotted line) or decreasing (light orange dashed line) scintillation efficiency extrapolations at low energy (Aprile et al. 2010), CoGeNT (+) and CRESST (Angloher et al. 2002) (blue dash-dotted line) (Figure 11 from Akerib et al. 2011)

based on the data from the most sensitive detectors, and Ge detectors dominated across the WIMP mass range, yielding a combined limit essentially equivalent to the Ge-only result (Akerib et al. 2011). Although the combined detector limit was designated as the primary outcome, the authors also reported Ge-only and Si-only limits. For low masses (< 6 GeV/c^2), the limit was set by Z2 3V and 6V intervals; for masses ≥ 10 GeV/c^2, contributions came from Z5 3V, Z5 6V, and Z2 3V data, while intermediate masses were dominated by Z3 3V (Akerib et al. 2011). Each Si detector contributed to the Si-only sensitivity, with Z6 providing the strongest limit below 4 GeV/c^2 and Z4 at higher masses. No serialized intervals spanned multiple detectors, confirming that serialization order does not affect the result (Akerib et al. 2011).

Smith et al. (2007) analyzed high-velocity halo stars from early RAVE survey data (Steinmetz et al. 2006), deriving a 90% confidence interval for the galactic escape velocity of 498–608 km/s, with a median of 544 km/s. While other experiments (as seen in Figures 13 and 14) adopt larger values—CRESST (Angloher et al. 2002) uses 650 km/s, and DAMA/LIBRA and CoGeNT employ 600 km/s—Akerib et al. (2011) adopts 544 km/s. Akerib et al. (2011) computed additional limits for 650 km/s to assess sensitivity. For WIMP masses > 4 GeV/c^2, differences are negligible; however, the Si-only limit at 544 km/s is up to 20% weaker between 6–10 GeV/c^2 (Akerib et al. 2011). The escape velocity assumption becomes critical below 4 GeV/c^2. Figure 9 compares limits for 544 km/s and 650 km/s down to 1 GeV/c^2, alongside the Smith et al. (2007)

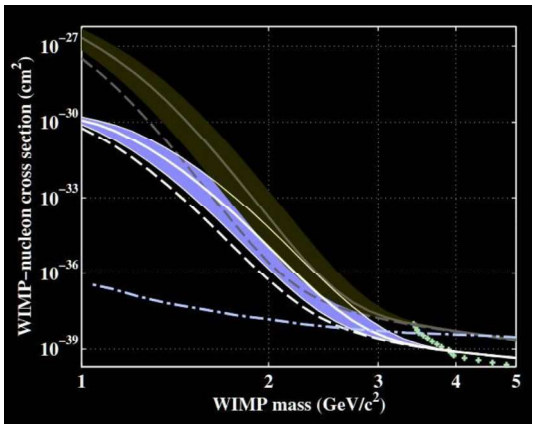

Figure 15: The 90% confidence level upper limits from the combined Ge and Si (white solid line) and the Si only CDMS shallow-site data (gray solid line) for WIMP masses as low as 1 GeV/c^2 with an assumed galactic escape velocity of 544 km/s as estimated by Smith et al. (2007). This figure compares these exclusion limits to those with an assumed escape velocity of 650 km/s as estimated by Donato et al. (1998) (black dashed line for combined data and gray dashed line for Si only data). The figure depicts the ranges of combined Ge and Si (blue shaded region) and Si only (yellow shaded region) limits corresponding to the 90% confidence level range of velocities found by Smith et al. (2007). The figure also depicts the limits from the CRESST (Angloher et al. 2002) (blue dash-dotted line, 650 km/s) and CoGeNT (+, 600 km/s) experiments (Figure 12 from Akerib et al. 2011).

confidence interval. The largest effect occurs at 2 GeV/c^2, where Ge and Si limits vary by over an order of magnitude and Si-only limits by more than two orders of magnitude (Akerib et al. 2011).

Akerib et al. (2011) determined detection efficiencies for hardware and software phonon thresholds and analysis cuts directly from data, with finite statistics introducing uncertainties. These uncertainties modestly affect sensitivity for WIMP masses below 6 GeV/c^2. For example, the combined Ge and Si exclusion limit at 5 GeV/c^2 is ~20% weaker when 90% confidence level lower-limit efficiencies (1.28σ below the mean) are used instead of best-fit values (Akerib et al. 2011). The effect is largest at 1 GeV/c^2, where the exclusion limit is reduced by a factor of five (Akerib et al. 2011). A conservative approach was therefore adopted, applying 90% confidence level lower-limit efficiencies in all exclusion limit calculations. Figure 5 of Akerib et al. (2011) compares best-fit and reduced efficiencies for representative Ge and Si detectors. Figures and tables preceding Figures 13–14 employ best-fit efficiencies, while Figures 13–14 are based on reduced values.

The authors evaluated the combined Ge and Si exclusion limit for sensitivity to uncertainties in Ge electron-recoil energy scales. YER-corrected and

Q-corrected recoil energies were checked across the 3V and 6V searches using calibration lines, with peak energies estimated via Gaussian fits derived from cesium-137 calibrations. Deviations were generally consistent with statistical fluctuations, though some detectors showed significant phonon overestimates at the 1.3 keV line (Akerib et al. 2011). The largest discrepancy occurred in Z2 6V data, with YER-corrected recoil energies overestimated by ~6% and Q-corrected values by ~16%, the latter reflecting combined ionization under-measurement and phonon over-measurement. The authors applied an ad hoc recalibration and recalculated the exclusion limits. Because discrepancies mainly involved recoil-energy overestimates, the recalibrated Ge and Si limit was marginally stronger, with visible differences only at the lowest WIMP masses. The effect peaked at 1 GeV/c^2, where the recalculated limit was ~2 times stronger. Final limits (Figs. 7–9) are based on the original Ge energy scales, a conservative choice avoiding systematic uncertainties from the recalibration.

4 Discussion

4.1 EDELWEISS

The results shown in Figure 4 of Benoit et al. (2002) are inconsistent with interpreting the modulation signal as arising from a WIMP described by the standard phenomenological model of Lewin and Smith (1996). In the energy range 20–64 keV, 9.8 nuclear recoils were expected, yet none were observed, corresponding to a Poisson probability of only 0.006%. The black triangle in Figure 4 at M_W = 44 GeV/c^2 and $\sigma_n = 5.4 \times 10^{-6}$ pb marks the value quoted by DAMA when combining their modulation data with limits obtained via pulse-shape discrimination in NaI (Bernabei et al. 1996). This point is also incompatible with the present EDELWEISS results, where the probability of observing zero events given a prediction of 6.2 is 0.2%. The only surviving portion of the 3σ DAMA NaI1-4 region corresponds to neutralino masses below the 45 GeV/c^2 bound established at LEP (Laktineh 2002). Thus, the standard framework of Lewin and Smith (1996) fails to reconcile the EDELWEISS and DAMA findings.

Figure 5 of Benoit et al. (2002) compares the combined EDELWEISS exclusion limit with those obtained from other direct WIMP searches. For spin-independent WIMP–nucleon interactions, the EDELWEISS sensitivity is currently the strongest achieved by any dark matter experiment for WIMP masses above 35 GeV/c^2. In addition, the EDELWEISS data begin to probe specific supersymmetric models that predict the largest interaction rates. As an illustration, Figure 5 includes the range of masses and cross-sections allowed in the calculations of Bottino et al. (2001) and Baltz and Gondolo (2001). The EDELWEISS exclusion limits allow a large portion of these ranges, suggesting some support for a supersymmetric dark matter candidate.

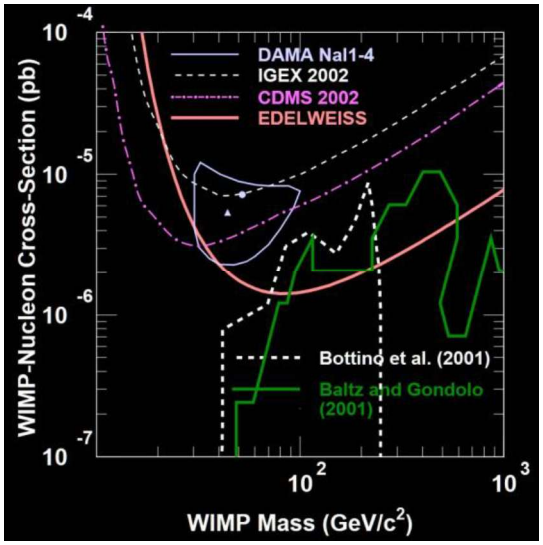

Figure 16: EDELWEISS exclusion limits compared with limits from other works with titles included in the figure (Figure 5 of Benoit et al. 2002).

4.2 CDMS

It is reasonable to suspect that the sensitivity to low-mass WIMPs could depend strongly on the lowest-energy trigger efficiency and on how the finite energy resolution near threshold is accounted for (Akerib et al. 2011). If so, even modest unknown systematics in hardware trigger efficiencies might significantly affect the exclusion limits. To test this, Akerib et al. (2011) examined the combined Ge and Si limit under two scenarios. First, the authors set the hardware trigger efficiency of each detector to zero wherever it fell below 10%, effectively removing the lowest-energy detection capability. The resulting limit was only slightly weaker than the full-efficiency case, with the largest difference—about 3%—occurring at a WIMP mass of 1 GeV/c^2. Second, the authors explored the impact of possible systematic uncertainty from near-threshold energy resolution by smearing the hardware trigger efficiencies by half of the YNR-corrected recoil-energy resolutions (~50 eV for Ge and ~100 eV for Si) before applying them in the limit calculation. This test addresses the concern that an additional resolution factor may be embedded in the efficiency estimates due to differences between the phonon signal used for triggering and the offline-reconstructed phonon energy. While deconvolution would be the more rigorous approach, the reverse operation is easier to implement and produces a comparable result. In practice, the resulting limit is slightly stronger, with the most significant difference—about 5%—again at 1 GeV/c^2. Overall, these studies indicate that the low-mass WIMP sensitivity for Run 21 of the CDMS experiment is fairly robust against systematic uncertainties in hardware trigger efficiencies. As shown in Figure 10 of Akerib et al. (2011), these robust upper limits appear

to support predictions for a supersymmetric dark matter candidate as described in Bottino et al. (2003).

4.3 Further Discussion

Cryogenic direct-detection experiments such as the EDELWEISS and CDMS experiments have made significant progress in constraining WIMP parameter space. The 2002 EDELWEISS run demonstrates how advances in detector technology can improve data-collection efficiency and reliability. Similarly, the deployment of ZIP detectors with phonon-mediated readout in Run 21 of the CDMS experiment extended sensitivity to recoil energies as low as about one keV.

The data from Run 21 of the CDMS experiment and the EDELWEISS 2002 run both suggest some support for supersymmetric dark matter candidates, as predicted by Bottino et al. (2001) and Bottino et al. (2003). The exclusion of portions of the DAMA modulation signal region by CDMS and the upper limits set by EDELWEISS below 10 TeV/c^2 provide valuable constraints on supersymmetric parameter space (Akerib et al. 2011, Benoit et al. 2002). The improved sensitivity to low-mass WIMPs strengthens the case for lighter supersymmetric particles. These exclusion limits offered by cryogenic detectors indicate a promising direction for the direct detection of dark matter.

The future of cryogenic detection is promising. Continued refinement of sensor technology, reduction of microphonic noise, and expansion of detector arrays will enable exposures on the order of hundreds of kg·d with thresholds approaching the sub-keV scale. Such advances will allow experiments to probe cross sections well below the current limits. In this regime, cryogenic detectors will not only test the viability of supersymmetric particles but also guide future theoretical models for dark matter.

Cryogenic detectors have been effective at excluding significant regions of parameter space and previous limits, and at demonstrating high sensitivity to low-mass WIMP candidates. Scaling exposures and refining resolution will further strengthen exclusion limits. As these experiments advance, they will likely establish or refute supersymmetry as a viable framework for particle dark matter.

5 Conclusion

This paper has traced the development of dark matter research from its historical origins to its current status as a central problem in astrophysics and particle physics. Historical analyses by Kelvin, Zwicky, Freeman, and Rubin established the observational foundation for the existence of non-luminous matter, with galaxy dynamics and cluster mass discrepancies providing compelling evidence for dark matter halos. This observational evidence motivated modern experimental methods that aim to directly detect particles predicted by particle models, including neutrinos, axions, and supersymmetric candidates. We then

examined the role of cryogenic direct-detection experiments, focusing on the EDELWEISS 2002 run and Run 21 of the CDMS II collaboration. Both experiments have demonstrated the effectiveness of cryogenic bolometers in achieving low recoil thresholds, improved baseline resolutions, and robust fiducial-volume definitions. The 2002 EDELWEISS run achieved an effective exposure of 7.4 kg·d and set upper limits on WIMP–nucleon scattering cross sections, while CDMS extended sensitivity to recoil energies as low as one keV and excluded portions of the DAMA modulation signal region. These results provide valuable constraints and suggest some support for previous predictions for supersymmetric dark matter. The demonstrated sensitivity of cryogenic detectors to low-mass WIMPs displays their importance in attempting to prove supersymmetric models and guiding theoretical development. As detector technology continues to improve, cryogenic experiments will remain at the forefront of direct detection, offering the potential to either confirm or exclude supersymmetry as a viable framework for particle dark matter.

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